shadows of their former selves, merely holding data; all the interesting code has moved to the DAOs and the components of your web application. The saddest part of this situation is that you can reuse almost none of the earlier code; the structure of the code has completely changed.

The “controllers” in your web application combine DAO methods to achieve changes in the data (`findConversations`, `insertMessage`, and so on). This combination of methods results in an interaction with the database that you can’t easily predict; the controllers are free to combine the database operations in any way, as illustrated in figure 1.6.

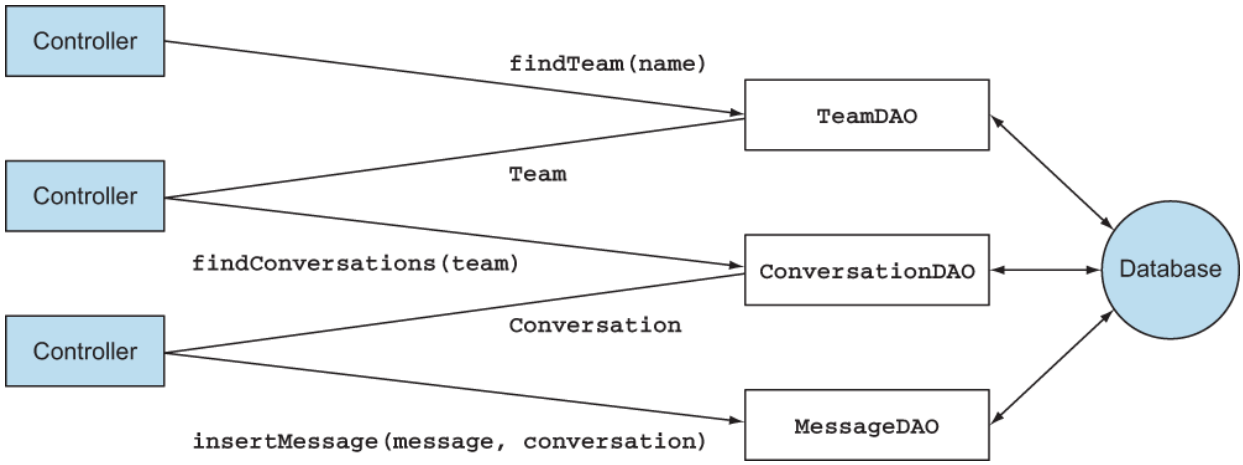


Figure 1.6 DAO interactions

The figure shows one possible flow through the code to add a message to a conversation. You can imagine numerous variations of database access flows using DAOs. Allowing any party to mutate or query records at any time can lead to performance trouble that you can’t predict, like deadlocks and other problems. It’s exactly the kind of complexity you want to avoid.